

Nicolas Alarcon

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EDUCATION

Columbia University (New York, NY) – *B.S. Electrical Engineering & Computer Science, M.S. E.E.*

Sept. 2021 - May 2025, Sept 2024 - May 2026 | GPA: 4.08

- **Honors:** C.P. Davis Scholar, HSF Scholar, Tau Beta Pi, Dean's List, Student Excellence Award in Electronics
- **Relevant Coursework:** Microwave Circuits, VLSI, Computer Architectures, Embedded Systems, Robotics Studio, Digital Signal Processing, Analog Electronic Circuits, Analog Filters, Computer Networks
- **Activities:** Columbia Bartending Agency Instructor, NicoMakes Etsy Shop Owner, NASA L'Space Academy

EXPERIENCE

Lelantos – *Electrical Engineering Intern*

May 2025 - August 2025 | New York City, NY

- Designed and verified a 0.1–4 GHz impedance-matched measurement PCB in KiCAD for a ppb resolution CO₂ sensor, using Keysight ADS to ensure RF performance across broadband frequencies
- Developed firmware for Teensy 4.x MCUs and a Python front end to enable variable sensor biasing, automated logging, and streamlined lab evaluation of over 150 sensor samples
- Led post-silicon bring-up and debugging of a mixed-signal ASIC I²C programming interface

Thomas J. Watson Research Lab, IBM – *Quantum Hardware Engineering Intern*

May 2024 - August 2024 | Yorktown Heights, NY

- Independently designed and executed a VNA Analysis Dashboard for decreasing time in diagnosing system failures utilizing statistical analysis techniques for comparison against historical data
- Mounted VNA Analysis Dashboard as secure Web Server for ease in large-scale analysis of 2-Port S-Parameters
- Developed Python device drivers for VNA and ZTM Switch Matrices to facilitate mass signal integrity analysis

The Makerspace, Columbia University – *Superuser Volunteer (2022), Student Administrator*

March 2022 - Present | New York, NY

- Provide design consultations and assistance in manufacturing products for research and personal projects including a turntable, rocket/car parts, high-resolution molds, electroplating, and architectural models
- Host project and training seminars in CAM/CAD in Fusion 360, Leatherworking, and Waterjet Manufacturing
- Source and create SOPs for new equipment including a kiln, blowtorch, wood lathe, and leather toolkit

PROJECTS

SystemVerilog RISC-V Out-of-Order Pentium6 Pipelined Processor

- Developed synthesizable, reduced-set RISC-V-32I ISA processor handling both C and ASM benchmarks
- Featured 256B Dcache, non-blocking 256B prefetched Icache, split LSQ, two-bit branch predictor for 2x improvement on original design. All verification conducted via custom Verilog, Assembly and C testbenches.
- Tools: RISC-V, SystemVerilog, C, Make, Git, Bash, Linux

Beamforming Audio to Subtitle Augmented Reality Glasses

- Designed communication aid projecting real-time subtitles onto smart glasses using a microphone array.
- Received First Place in Electrical Engineering Senior Design Expo
- *Tools:* NLP, Analog Circuit/PCB Design, Matlab, Python, MCU Development, Raspberry Pi

CycloneV De1 SoC FPGA GameBoy Hardware Emulator

- Created a hardware emulator accurate to patented Nintendo schematics integrated within a larger IP with modern technological upgrades. Developed automated Verilator testbenches for incremental development.
- *Tools:* FPGA, SystemVerilog, Logic Design, Embedded Systems, C, C++, Quartus, Verilator, Linux

SKILLS & INTERESTS

- *Software:* Cadence, Spice, Fusion, KiCad, Python, Verilog, Java, Git, Matlab, C/C++, Linux, MS Office, AutoCAD
- *Technical:* Soldering, Circuit Analysis, Electronics, 3D Printing, Laser Cutting, Prototyping, PCB Manufacturing
- *Languages:* Spanish, French, English
- *Interests:* Volleyball, Art History, Bartending, Woodworking, Cooking, Fencing